METRIC

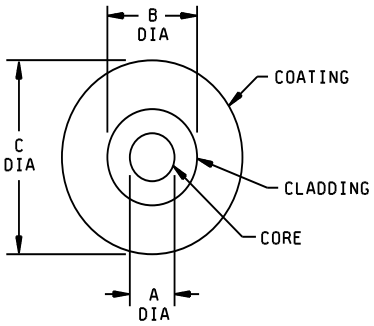
MIL-PRF-49291/10B
w/AMENDMENT 2
17 July 2019
SUPERSEDING
MIL-PRF-49291/10B
w/AMENDMENT 1
19 July 2016

PERFORMANCE SPECIFICATION SHEET

# FIBER, OPTICAL, TYPE I, CLASS I, SIZE IV, COMPOSITION A, WAVELENGTH B, RADIATION RESISTANT, ENHANCED PERFORMANCE CHARACTERISTICS (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-49291.



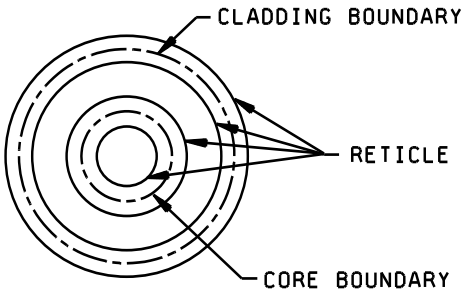

| PIN | Dimensions | | |
|---|---|---|---|
| | A (µm) | B (µm) | C (µm) |
| M49291/10-01<br>M49291/10-01A | 62.5 ± 2.5 | 1/ | 125 ± 1 2/ |
| M49291/10-02<br>M49291/10-02A | 62.5 ± 2.5 | 125 ± 1 | 245 ± 10 3/ |

1/ Cladding diameter remains unspecified with condition that optical fiber must meet specified requirements.
2/ Overall dimension includes a non-strippable coating (or surface treatment). Cladding along with this coating shall conform to C diameter. Use C dimension in lieu of cladding to determine cladding ovality, core-to-cladding offset and cladding dimensions.
3/ Integrated fiber and cable manufacturing process may exceed this diameter. When this condition occurs, then this configuration shall conform to cabling specification in addition to this specification sheet.

FIGURE 1. Dimensions and configuration of optical fiber construction.



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| Circle (solid) | Diameter (µm) |
|---|---|
| Inner | 60.0 |
| Second | 65.0 |
| Third | 124.0 |
| Fourth | 126.0 |

FIGURE 2. Tolerance fields.

Part or identifying number (PIN). (See figure 1 and table II):

M49291/10-01
M49291/10-01A
M49291/10-02
M49291/10-02A

REQUIREMENTS:

Dimensions and configuration:

Diameter: See figure 1 and figure 2. (Diameter requirements are based on selecting fibers with end-point tolerances of 125 ± 1 µm from production with a 125 ± 2 µm tolerance.)

Ovality:
Core: ≤ 5 percent.
Cladding: ≤ 1 percent.

Offset:

Core-to-cladding: ≤ 2 µm

Fiber-to-coating:
Coating-cladding concentricity error: ≤ 10.5 µm.
Overall coating concentricity ratio (OCCR or coating noncircularity): ≥0.75 for a 250 µm diameter coating.

Fiber curl: Not applicable.

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Splices within the optical fiber: Splices within the optical fiber are not allowed.

Proof test: 1.38 GPa (200 kpsi).

Fiber mass/unit length: 0.1 kg/km maximum.

Optical parameters:

Change in optical transmittance: Measurements to be made at 850 nm ±20 nm. Multiple fibers may not be concatenated during the measurement of change in optical transmittance. The center wavelength and light launch conditions for the specified fiber size and temperature range shall be as specified in MIL-STD-1678-2.

Insertion loss: The insertion loss measurements shall be performed per TIA-455-34 with the exception for adhering strictly to the setup and test procedure specified in Measurement 2101 of MIL-STD-1678-2. Multiple fibers may not be concatenated during the measurement of change in optical transmittance or optical discontinuity. The center wavelength and light launch conditions for the specified fiber size and temperature range shall be as specified in MIL-STD-1678-2.

Maximum attenuation rate: Spectral attenuation shall be performed at a sufficient number of wavelengths to cover the applicable optical transmission windows and sources for attenuation. Sufficient coverage includes the optical transmission windows (850 nm, 1,300 nm), ultraviolet absorption band (600 nm to 870 nm), Rayleigh scattering losses (800 nm to 1,200 nm), OH impurity absorption peaks (945 nm, 1,249 nm, and 1,383 nm). Use of a source and power meter to perform this measurement is not acceptable. Attenuation rate for wavelengths of operation shall not exceed the maximum values listed below.

Maximum attenuation rate over the wavelength range from 800 nm to 1,600 nm shall be less than 4.5 dB/km. Attenuation rate for wavelengths of operation shall not exceed 3.0 dB/km at 850 nm $\pm$ 20 nm and 0.7 dB/km at 1,300 nm $\pm$ 20 nm.

Numerical aperture: 0.275 $\pm$ 0.015 at 850 nm $\pm$ 25 nm.

Overfill bandwidth: ≥ 300 MHz-km at 850 nm $\pm$ 20 nm and ≥ 600 MHz-km at 1,300 nm $\pm$ 20 nm.

RML/$EMB_c$ bandwidth: ≥ 385 MHz-km at 850 nm $\pm$ 20 nm and ≥ 700 MHz-km at 1,300 nm $\pm$ 20 nm.

Chromatic dispersion: Zero dispersion wavelength shall be 1,343 nm ± 23 nm. The value of the slope at the zero dispersion wavelength shall be not greater than 0.11 ps/$nm^2$-km for a zero dispersion wavelength between 1,320 nm to 1,348 nm and shall not be greater than 0.001 (1,458-zero dispersion wavelength) ps/$nm^2$-km for a zero dispersion wavelength between 1,348 nm to 1,366 nm.

Transient attenuation: Transient attenuation shall be ≤ 1.5 dB at 1,300 nm when tested in accordance with TIA-455-78.

Attenuation uniformity: Applicable except perform at 850 nm and 1,300 nm. There shall be no discontinuities in attenuation along the length greater than 0.2 dB.

Macrobend attenuation: Applicable except test shall be performed in accordance with TIA-455-62, method B, with optical power measurements obtained for each row of parameters in table I.


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TABLE I. Macrobend attenuation test parameters.

| Mandrel diameter mm (inch) | Number of turns around mandrel | Measurement wavelength (nm) | Macrobend attenuation (dB) |
|---|---|---|---|
| 76 ± 1 (3.0 ±.04) | 100 | 1300 ± 20 | 0.5 |
| 76 ± 1 (3.0 ±.04) | 100 | 850 ± 25 | 0.5 |
| 30 ± 1 (1.2 ± .04) | 10 | 1,300 ± 20 | 1.5 |
| 30 ± 1 (1.2 ± .04) | 10 | 850 ± 25 | 1.5 |
| 20 ± 1 (.79 ± .04) | 1 | 1300 ± 20 | 2.5 |
| 20 ± 1 (.79 ± .04) | 1 | 850 ± 25 | 2.5 |
| 15 ± .5 (.59 ± .02) | 1 | 1,300 ± 20 | 3.5 |
| 15 ± .5 (.59 ± .02) | 1 | 850 ± 25 | 3.5 |

Polarization mode dispersion: Not applicable.

ENVIRONMENTAL:

Temperature range: See table II.

TABLE II. Temperature range. 1/

| PIN | Operating °C (°F) | Nonoperating °C (°F) | Storage °C (°F) |
|---|---|---|---|
| M49291/10-01<br>M49291/10-02<br>2/ | -55 to +165 (-67 to +329) | -55 to + 85 (-67 to +185) | -55 to +85 (-67 to +185) |
| M49291/10-01A<br>M49291/10-02A<br>3/ | -46 to +85 (-51 to +185) | -55 to +85 (-67 to +185) | -55 to +85 (-67 to +185) |

1/ Temperatures are specified in degrees Centigrade, Fahrenheit equivalents are specified in parentheses.
2/ Temperature ranges in row 2 conform to those specified for other MIL-SPEC fiber optic cabling components for aircraft applications.
3/ Temperature ranges in row 3 conform to those specified for other MIL-SPEC fiber optic cabling components for standard applications (such as the telecommunication industry).


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Fluid immersion aging: Not applicable.

Mechanical strippability: Applicable except, in addition, each individual optical fiber shall comply with the below requirements and test method for diameter in addition to the 3.5.3 (of MIL-PRF-49291) residue coating and fiber damage. Also, guidance is provided for determining the strip force using the method specified in TIA-455-178.

Strip force determination. When the mechanical strippability is performed to TIA-455-178, the test apparatus (such as a tensile test machine) parameters should be set up for adequate mechanical strippability to measure the strip force. Stripping speed (rate) and duration may be determined and set to achieve the strip force over an approximate length. Guidance for starting values to determine stripping speed and time can be obtained in 5.2 of TIA-455-178 for stripping rate and 4.1 of TIA-455-178 for length and iterate as needed. A plot of strip force versus time (or strip length) for each strip performed is preferred. Ten tests for strip force shall be performed on each length of optical fiber length cut from each spool.

Diameter requirement. Requirement that the stripped optical fiber complies with diameter B in figure 1 shall be verified by successful insertion and removal of each optical fiber stripped end into a ring gauge/die/ferrule with a hole diameter of 125 +1/-0 micron. It is in the interest of the implementer to choose a gauge closer to 126 micron than 125 microns.

Coating adherence. Requirement for coating adherence shall be verified by the performance of a fiber pullout test. The end of each stripped length of optical fiber shall be terminated onto the end of a MIL-PRF-29504 terminus (or other suitable fiber optic connector if approved by the qualifying activity prior to test sample fabrication). Fabrication shall be performed as specified in Requirement 4101, Requirement 4102 or Requirement 4103 of MIL-STD-1678-4, as applicable. The fiber pull out force shall be tested by applying an axial tensile load between the optical fiber and the terminus for a minimum duration of one minute. For M49291/10-01 and M49291/10-02, this tensile load shall be 3.0 N (.67 pounds). For M49291/10-01A and M49291/10-02A, this axial tensile load shall be 14.0 N (3.1 pounds). At the completion of the test, the optical fiber shall be visually examined and length measured to verify that no fiber pull out occurred. One optical fiber length from each spool used for the Group 2 mechanical strippability test shall be tested for fiber pullout force. The test entity is given the latitude to affix the terminus in a clamp, corrector or other fixture of their choice and the method of applying the force.

Splice compatibility. Six 10 meter (32.8 feet) lengths of optical fiber conforming to the requirements of this specification sheet shall be spliced, (using either a fusion splicer or mechanically splice) and connectorized (using MIL-PRF-29504/4 and MIL-PRF-29504/5 termini) to optical fiber conforming to MIL-PRF-49291/6. Three sample lengths are to be used for each splice or connector. Start by obtaining the baseline measurement of the uncut, 10 meter length, of optical fiber. Cut the optical fiber in half (i.e., at 5 m from the end) and either terminate each fiber (at the 5 m cut) with MIL-PRF-29504/4 and /5 termini or splice a MIL-PRF-49291/6 optical fiber between the cut ends. An insertion loss measurement shall be performed in accordance with Method A2 (i.e., a restricted launch condition) of TIA-455-34. The insertion loss shall not exceed 0.2 dB per splice and 0.5 dB per connector mated pair when performed at a wavelength of 1,310 nm. Repeat this sequence until three 10 meter lengths is terminated with termini in the middle and three 10 meter lengths are spiced with MIL-PRF-49291/6 optical fiber in the middle.

Dynamic tensile strength: Applicable except that 30 test samples from each spool shall be tested with no humidity (i.e., under specified room conditions) with the calculations performed for the 30 samples tested. Testing shall be performed with the gage length of 500 mm (19.7 inch), test speed (strain rate) of 2.5 mm/min (0.098 in/min) and speed to pre-load of 1.25 mm/min (0.049 in/min).

Fungus resistance: Fungus resistance is applicable.


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Thermal shock:

Applicable as specified for M49291/10-01A and M49291/10-02A.

Applicable for M49291/10-01 and M49291/10-02, except that the temperatures of -55°C +0/-5°C (-67°F +0/-8°F) and 165°C +5/-0°C (329°F +8/-0°F) shall be used for the low and high soak temperatures, respectively. Test shall be performed in accordance with TIA-455-71, schedule C-0 (5 cycles). The change in optical transmittance shall be measured during (towards the end of each soak temperature) and after the test.

Storage temperature: Applicable for M49291/10-01A and M49291/10-02A. Not applicable for M49291/10-01 and M49291/10-02.

Temperature cycling:

Applicable as specified for M49291/10-01A and M49291/10-02A.

Applicable for M49291/10-01 and M49291/10-02 except for the optical fiber with a -55°C to 165°C (-67°F to 329°F) temperature range, perform the following in lieu of the specified temperature cycling test. Optical fiber shall be tested in accordance with TIA-455-3 using the setup and procedure specified in Measurement 3301 of MIL-STD-1678-3 with the temperature profile in table 3301-II. The change in optical transmittance shall be measured during and after the test. At a minimum for the “during test” measurements, an optical transmittance measurement shall be performed towards the end of each soak period (maintain step) after every 50 cycles. A post test visual examination of the test specimens shall reveal no surface impairment or any damage detrimental to the operation of the test specimens. Test fixtures, if used, must be of minimum mass and approved by the qualifying activity. No other mass (item that causes significant thermal lag) shall be added inside the chamber.

Temperature life (life aging):

Applicable as specified for M49291/10-01A and M49291/10-02A.

Applicable for M49291/10-01 and M49291/10-02 except that the high exposure temperature shall be 165°C +5/-0°C (329°F +8/-0°F). Test samples shall be tested in accordance with TIA/EIA-455-4 for the duration of 1,000 hours at the high exposure temperature. The change in optical transmittance shall be measured after the test. An optical transmittance measurement is requested to be taken daily for informational purposes only. The optical fiber shall be visually examined after the test to the extent feasible. Optical fiber shall not be damaged, and there shall be no distortion, cracking of coating or other damage detrimental to the operation of the optical fiber. The requirement for the change in optical transmittance shall be met after the test. Performance of a cable/fiber retention test is not required as part of the final inspection for this test.

Nuclear radiation resistance: Nuclear radiation resistance requirements and test conditions shall be as shown below and in table III and table IV:

Light launch conditions: In accordance with TIA-455-78.

Wavelength: 1,300 nm ± 25 nm.

Source type: LED with FWHM spectral width ≤ 160 nm.



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TABLE III. Steady state gamma radiation test conditions.

| Test temperature (°C) | Dose rate | Total dose (rad Si)) |
|---|---|---|
| -28 ± 2<br>25 ± 2<br>85 ± 2 | 50 +0, -20 rad (Si)/sec | Classified |

(Some nuclear radiation resistance characteristics of this optical fiber are classified and shall be obtained from the qualifying activity. Application to receive these requirements must be made through the Department of the Navy, Naval Surface Warfare Center, Dahlgren Division, ATTN: Code B35, 17320 Dahlgren Road, Dahlgren, VA 22448-5100. Information concerning security clearance classification and "need to know" must be detailed in the request.)

TABLE IV. Steady state gamma radiation test requirements.

| Maximum induced attenuation (dB/km) | Attenuation at specified recovery time (dB/km) | Specified recovery time (sec) |
|---|---|---|
| ≤ 50 1/ | ≤ 15 at -28°C (-18°F) 1/<br>≤ 5 at 25°C(77°F)<br>≤ 5 at 85°C (185°F) | 1,000 |

1/ The radiation induced loss for a given threat. The total dose associate with the threat is classified and not necessarily equal to the test total dose.

CONFORMANCE:

Test samples for qualification: Reels of optical fiber produced for initial qualification and quality conformance shall be from at least two different preforms (see 4.5.1 of MIL-PRF-49291).

Qualification: Qualification shall consist of performing testing specified in table V.

In group A testing length may be measured using mechanical methods.

In group C testing the mechanical strippability test may be omitted if the optical fiber coatings have not changed from when the mechanical strippability test was last performed. The manufacturer shall provide a certificate of compliance for mechanical strippability in the group C test report.


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TABLE V. Qualification inspections.

| Test performed | Initial qualification |
|---|---|
| Initial uncoiling preparation | |
| Group 1 (3 spools-2km each) 1/ | |
| Visual and mechanical | X |
| OTDR tests: | |
| Fiber length | X |
| Attenuation uniformity | X |
| Parameter tests: | |
| Attenuation rate | X |
| Numerical aperture (MM only) | X |
| Core diameter (MM only) | X |
| Cutoff wavelength(SM only) | |
| Mode field diameter (SM only) | |
| Transient attenuation (MM only) | X |
| Macrobend attenuation | X |
| Bandwidth-overfill (MM only) | X |
| Bandwidth-RML/EMBc (MM only) | X |
| Chromatic dispersion | X |
| Polarization mode dispersion (SM only) | |
| Fiber curl (SM only) | |
| Mass/unit length | X |
| Geometry tests: | |
| End view: | |
| Core ovality (MM only) | X |
| Cladding diameter | X |
| Cladding ovality | X |
| Core to cladding offset | X |
| Side view: | |
| Coating diameter | X |
| Coating-cladding concentricity error | X |
| Overall coating concentricity ratio | X |



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TABLE V. Qualification inspections Continued.

| Test performed | Initial qualification |
|---|---|
| Group 2(samples off 3 spools) 1/ | |
| Tensile proof | X |
| Mechanical strippability | X |
| Dynamic tensile strength | X |
| Splice compatibility | X |
| Coating adherence | X |
| Group 3(same 3 spools-2km each 1/, 2/, 5/, 6/) | |
| Thermal shock 4/ | X |
| Storage temperature 4/ | X |
| Temperature humidity cycling 4/ | X |
| Mechanical strippability (post TH cycling) | X |
| Temperature cycling 4/ | X |
| Temperature life (life aging) 3/ , 7/ | X |
| Mechanical strippability (post Temp life) | X |
| Group 4(samples off 3 spools) | |
| Nuclear radiation resistance 4/ | X |
| Fungus resistance | X |

Notes:

1/ See table I in basic specification (MIL-PRF-49291) for specimen lengths.

2/ OOT = Change in optical transmittance. Perform in accordance with TIA-455-20 for transmitted power adhering strictly to the setup and test procedure specified in Measurement 2102 of MIL-STD-1678-2.

3/ Perform OOT after the test only.

4/ Perform OOT both during and after the test.

5/ Sample lengths: Recommend to prepare before Group 1. For mechanical strippability, place in pans and remove from chamber after specified environmental test.
Mechanical strippability: each spool, 2 samples, 91.4 cm (3 ft) long.
Mechanical strippability-post temp/humidity: each spool, 2 samples, 91.4 cm (3 ft) long.
Mechanical strippability-post life aging: each spool, 2 samples, 91.4 cm (3 ft) long.
Dynamic tensile strength: each spool, 30 samples, 198.1 cm (6.5 ft) long (no humidity).
Nuclear radiation resistance: each spool, 1 sample, 200 meters (656.1 ft) long on spool.
Fungus resistance: each spool, 1 sample, 30.5 cm (1 ft) long.

6/ Prepare in loose coils and place in pans. Recommend to terminate with single ferrule connectors (such as ST connectors) on the fiber ends (for attachment to test instrumentation optical ports – to obtain more stabilized measurements).

7/ Measurements requested daily for informational purposes.



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Qualification by similarity:

Change in glass:

Manufacturers who are qualified under this specification sheet and whose optical fiber with a change in the glass (composition or profile) passes the tests specified in table VI are qualified under this specification sheet for the optical fiber with changed glass.

TABLE VI. Change in glass.

| |
|---|
| Visual and mechanical |
| Fiber length |
| Attenuation uniformity |
| Attenuation rate |
| Numerical aperture |
| Core diameter |
| Transient attenuation |
| Macrobend attenuation |
| Bandwidth |
| Chromatic dispersion |
| Temperature cycling |
| Nuclear radiation resistance |

Change in coating:

Manufacturers who are qualified under this specification sheet and whose optical fiber with a change in the coating (composition or thickness) passes the tests specified in table VII, are qualified under this specification sheet for the optical fiber with changed coating.



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TABLE VII. Change in coating.

| |
|---|
| Visual and mechanical |
| Fiber length |
| Attenuation rate |
| Transient attenuation |
| Macrobend attenuation |
| Coating diameter |
| Coating noncircularity |
| Mechanical strippability |
| Dynamic tensile strength |
| Thermal shock |
| Storage temperature |
| Temperature humidity cycling |
| Temperature cycling |
| Temperature life (life aging) |
| Fungus resistance |

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: In addition to MIL-PRF-49291, this specification sheet references the following documents:

| | | |
|---|---|---|
| MIL-STD-1678-2 | TIA-455-3 | TIA-455-62 |
| MIL-PRF-29504 | TIA-455-4 | TIA-455-71 |
| MIL-PRF-29504/4 | TIA-455-20 | TIA-455-78 |
| MIL-PRF-29504/5 | TIA-455-34 | MIL-STD-1678-3 |
| MIL-PRF-49291/6 | TIA-455-178 | MIL-STD-1678-4 |


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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force – 85
DLA – CC
NASA - NA

Review activities:
Navy - SH
Air Force – 02, 19, 33, 93
DIA - DI

Preparing activity:
DLA - CC

(Project 6010-2019-001)

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